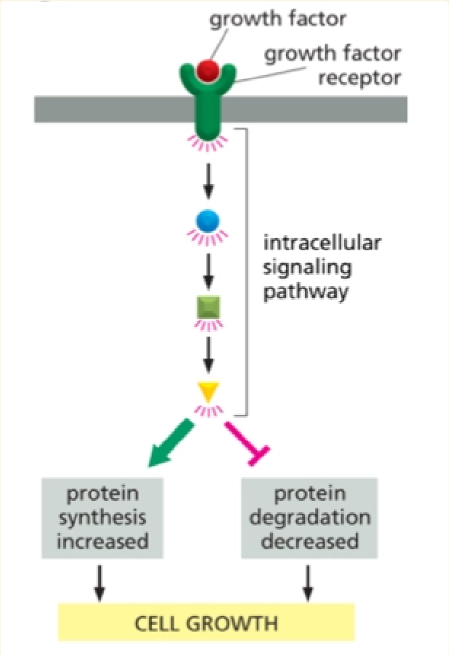
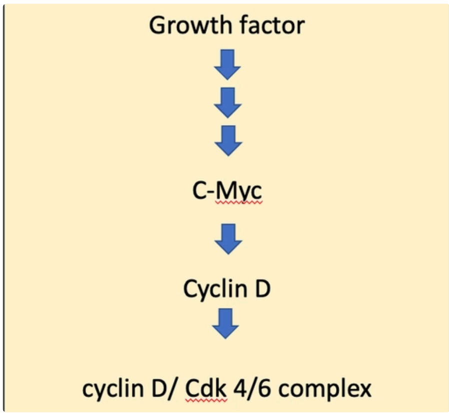
2.8 Cell Replication

- Give a reason why different cells divide at different rates.
    - Necessity for renewal (i.e some tissues do not need to be replenished as frequently as others)
        - Intestinal epithelial cells ~ 20 hours
        - Hepatocytes ~ 1 year
- Why is it that some cells never divide?
    - Some cells (i.e neurons and cardiac myocytes) never divide because they are **replenished from the differentiation of progenitor cells** and not from cell division
- Recall the order of the steps in the cell cycle.
    - G0 (quiescent phase)
    - G(ap)1 —> S —> G2 (interphase)
    - M(itosis)
- What happens in the quiescent (G0) phase?
    - A stage where the cell takes itself out of the cell cycle
    - The **cells are not dormant**, but **they are performing their functions without replicating**
- When does a cell go into the quiescent (G0) phase?
    - In the absence of a stimulus, a cell **stops dividing** and enters the quiescent (G0) phase
    - Cells are not dormant in this phase, but they are simply **not dividing** (i.e **they continue to perform their functions**)
- What is the purpose of the G1 phase in the cell cycle?
    - It is where **cells get bigger** and **increase the size of their proteins and organelles**
- What is the purpose of the S phase in the cell cycle?
    - **DNA replication**, to prepare it for cell division
- What is the purpose of the G2 phase in the cell cycle?
    - To check the **DNA is ready for cell division** in mitosis

- What stops a cell from going from the G1 phase to the S phase?
    - Lack of **nutrients**
    - Lack of **growth factors**
    - This prevents the cell growing to the required size
- What stops a cell from going from the G2 phase to the M phase?
    - DNA damage beyond repair
    - Incomplete DNA replication in S phase
- How does a cell decide to go from the G0 to the G1 phase?
    - **Growth factors stimulate** the entry of cells from the G0 to the G1 phase
        - This is the stimulus required to move a cell out of the G0 phase
    - The **growth factor binds onto the growth factor receptor**
    - This leads to a **cascade of phosphorylation reactions** which **results** in **increased protein synthesis** or **decreased protein degradation**
    - The cell is then ready to make the transition from the **G0 to the G1 phase.**



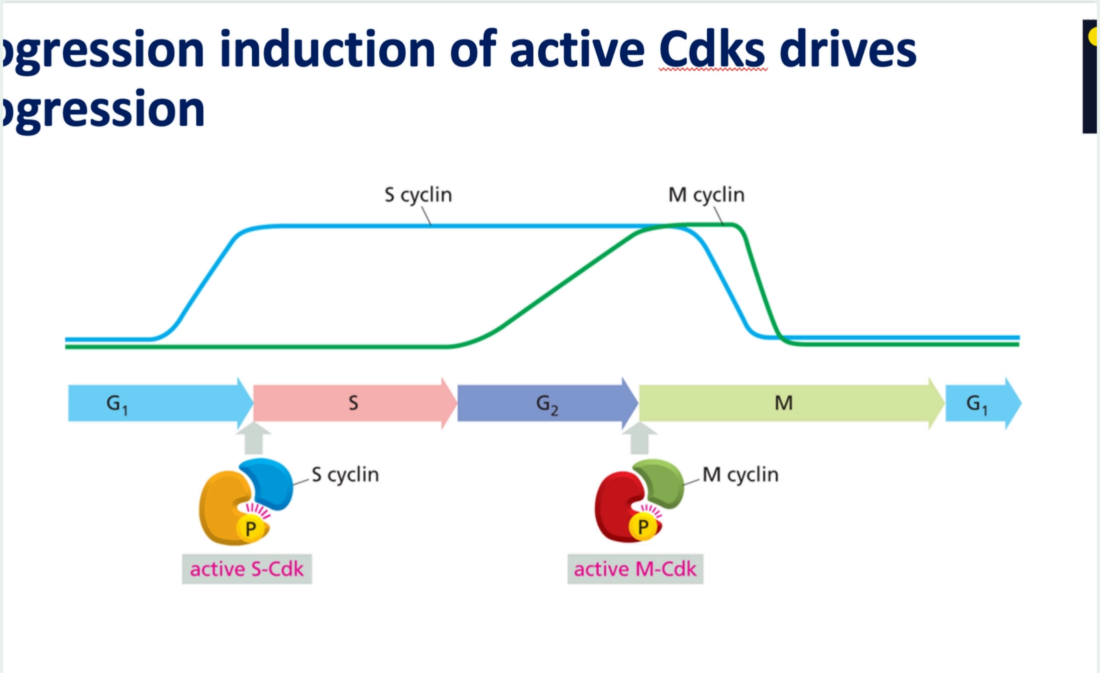
- Describe how growth factors interact with transcription factors.
    - **Growth factors bind onto the respective growth factor receptor** on the cell surface
    - This leads to a **cascade of phosphorylation reactions** which ultimately can lead to the **increase of certain transcription factors** and hence **altered gene expression**
        - This is because **transcription factors change the rate of transcription** by **controlling which genes are expressed** and **which are not expressed**
        - i.e transcription factors can turn on genes for DNA polymerase, thymidine kinase, cyclins etc.
    - This can cause increased growth in the cell
- What is c-Myc?
    - It is a **transcription factor** that promotes the **transition** of a cell from the **G0 to the G1 phase** of the cell cycle
        - It does this by **increasing the expression of genes that promote progression of the cell cycle**
    - It is used to **produce cyclins**
- When is c-Myc typically expressed?
    - When the **growth factors bind onto the growth factor receptors** on cell surfaces, and the **signalling cascade** of reactions begin to eventually express c-Myc.
- Diagram showing role of c-Myc in the cell cycle.



- What do the amino acids serine, threonine and tyrosine have in common?
    - They are all amino acids with **-OH groups**, making them **susceptible to phosphorylation**

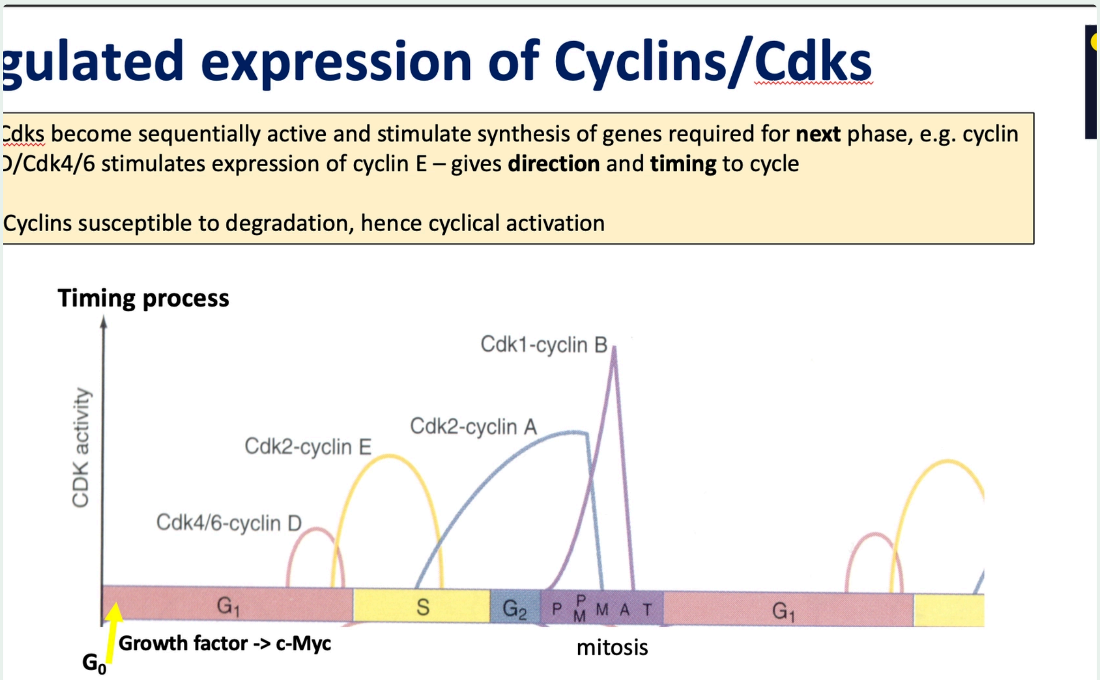
▼ What are cyclin-dependent kinases (CDKs)?

- They are **enzymes** that control the **progression of the cell cycle into relevant phases**
- **Different CDKs** are used to **progress to different stages of the cell cycle**
  - i.e **S-CDK** is used to move from **G1 to S phase** | **M-CDK** is used to move from **G2 to M phase**
- They are only **active when bound to cyclin**, and different cyclins are produced at different stages of the cell cycle, when needed (see below)



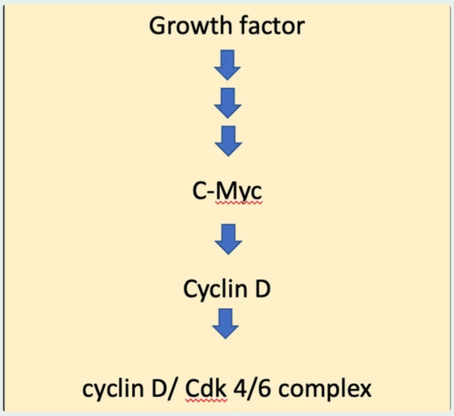
▼ Why are cyclins given their name?

- Their **concentrations fluctuate** at different levels throughout the cell cycle **via production and degradation** of the protein
- This important in **controlling the different stages** within the cell cycle.
- For example, **M-cyclin** is produced in build-up to **mitosis**, and **S-cyclin** is produced in the build-up to the **S phase.**
- Different cyclins can also induce the production of the next cyclin required in the cell cycle (see below)



▼ What is cyclin D?

- **Cyclin D is a cyclin** produced after expression of **the c-Myc transcription factor** caused by **growth factors and their resulting signalling cascades**
  - C-Myc turns on expression of genes that induce progression of the cell cycle, and **Cyclin D is produced** by expression of these genes in the **G1/S phase transition**

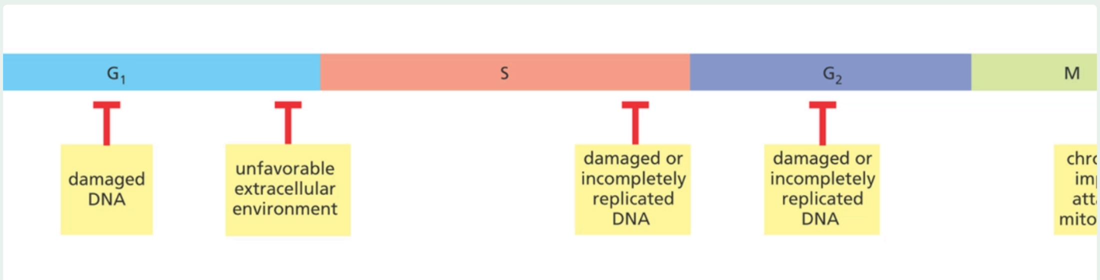


▼ What is the role of cyclin D?

- **Cyclin D** enables the **progression of the cell cycle** from the **G1 phase to the S phase** by binding onto its cyclin dependent kinase and **forming the Cyclin D: Cdk 4/6 complex**

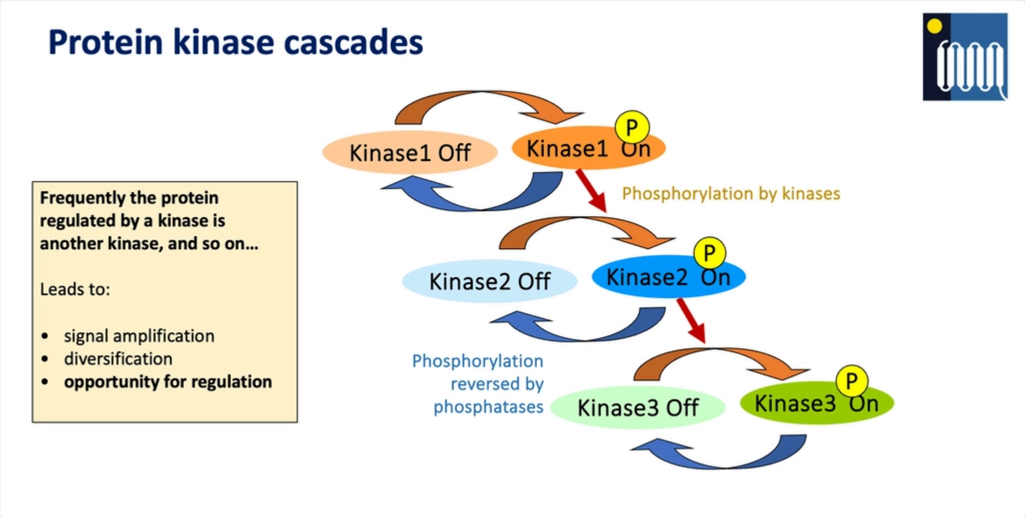
▼ What stops a cell progressing past the M phase of the cell cycle?

- Chromosomes being **improperly attached to the mitotic spindle**



▼ What is the protein kinase cascade?

- It describes how the activation of a kinase can lead to further activation of other kinases because the former kinase can phosphorylate the latter kinase
- This works because **kinases** are often involved in the **regulation of other kinases**



▼ What two processes does a cyclin-dependent kinase need to function?

- **Binding** (interaction) **with a cyclin**
- **Activation** (which occurs **via phosphorylation**)

▼ What is the name of the group of enzymes involved in dephosphorylation of substrates?

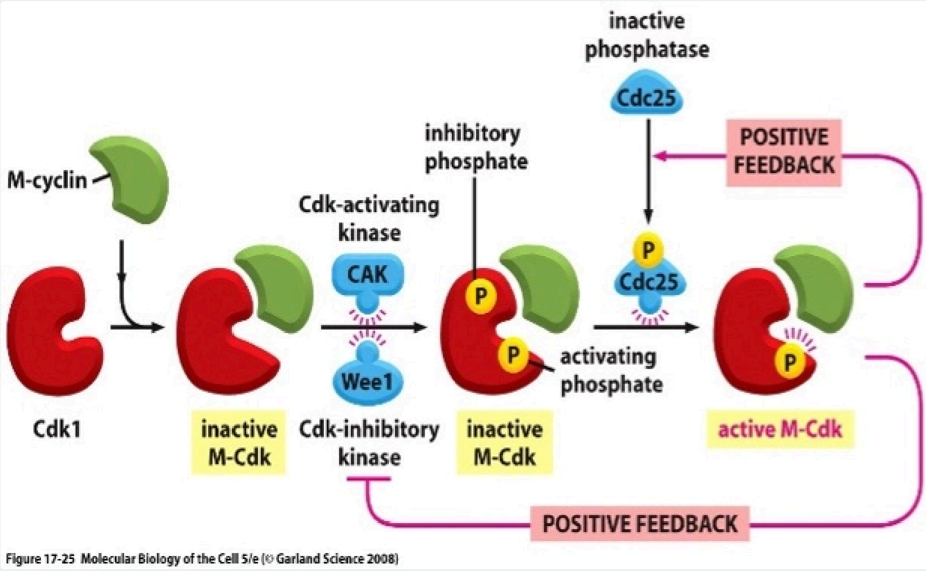
- **Phosphatases**
  - i.e **glucose-6-phosphatase** found in the liver to reverse action of Hexokinase IV

▼ Describe the difference between the expression of cyclins and cyclin-dependent kinases during the cell cycle.

- **Cyclin-dependent kinases (CDKs)** are expressed constantly **throughout the cell cycle**
  - However, they are not activated until cyclins bind to them and they are phosphorylated
- **Cyclins are expressed transiently** at different levels throughout the cell cycle
  - They are **constantly synthesised and degraded**

▼ Describe the process through which CDKs are activated.

- **CDK is bound to a cyclin** which is produced via **transcription factor** which is expressed via a **growth factor** which has bound onto the **growth factor receptor** on the cell surface.
- The CDK is now **phosphorylated** at both the **inhibitory and activatory sites** by **protein kinases**
- The CDK is then **dephosphorylated at the inhibitory site** by **activating phosphatases** in order to become **activated**.



▼ Describe how positive feedback within the CDK mechanism drives the cell cycle forward.

- As CDKs are activated, positive feedback leads to the **phosphorylation** (and hence activation) of **inactive phosphatases** and likewise **activation of inhibitory kinases**
- This means the CDK is continually phosphorylated and dephosphorylated and it remains active, allowing the CDK itself to phosphorylate other molecules (**perform its role**)
- The cell cycle hence is driven forward.

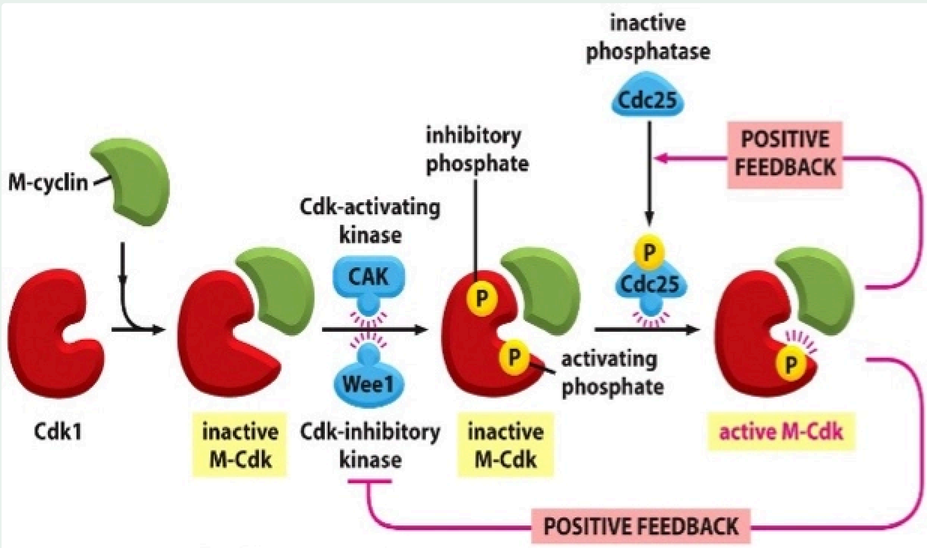
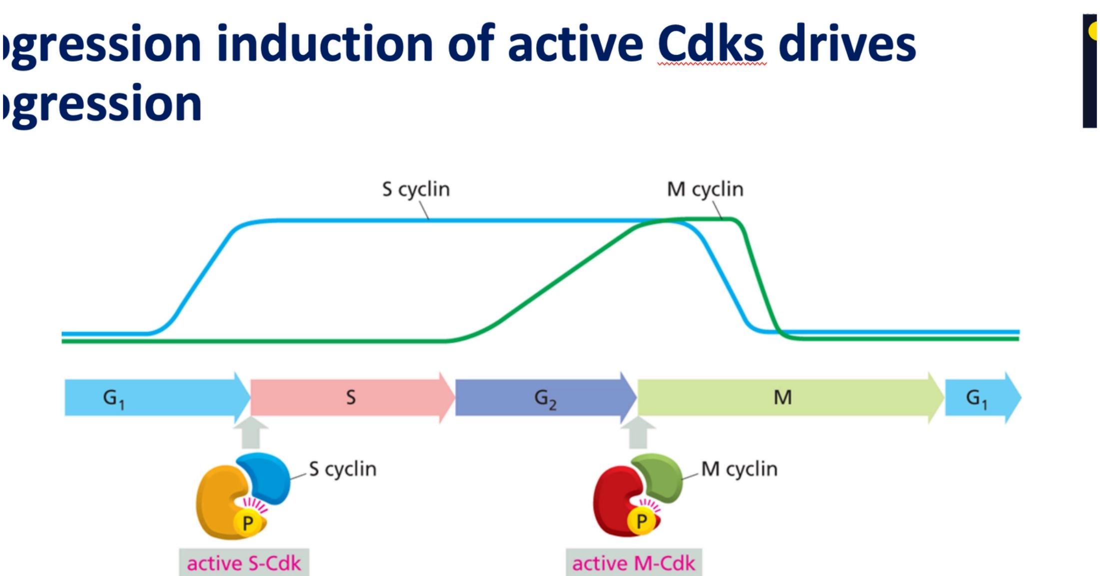


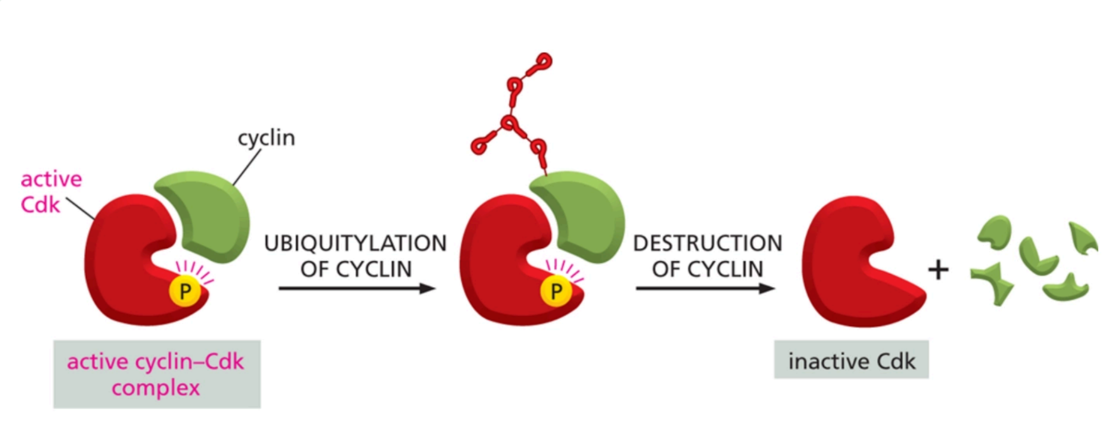
Figure 17-25 Molecular Biology of the Cell 5/e (© Garland Science 2008)

▼ Diagram showing how induction and degradation of different cyclins at various times drives the cell cycle forward.



▼ How are cyclins degraded?

- **Ubiquitination**
- A molecule called a **ubiquitin** is **added** to a residue on the **cyclin**, allowing the **proteasome to degrade the cyclin**
- This causes the cyclin to break down and in turn, **deactivate the CDK,** in a process called **proteasomal degradation**



▼ What is the function of thymidine kinase?

- Thymidine kinases have a key function in the **synthesis of DNA and therefore in cell division**
- They work alongside **DNA polymerases**
- Specifically, **thymidine kinase** is an **enzyme that creates a nucleotide by phosphorylating nucleoside**.

▼ What is a retinoblastoma and where is it found?

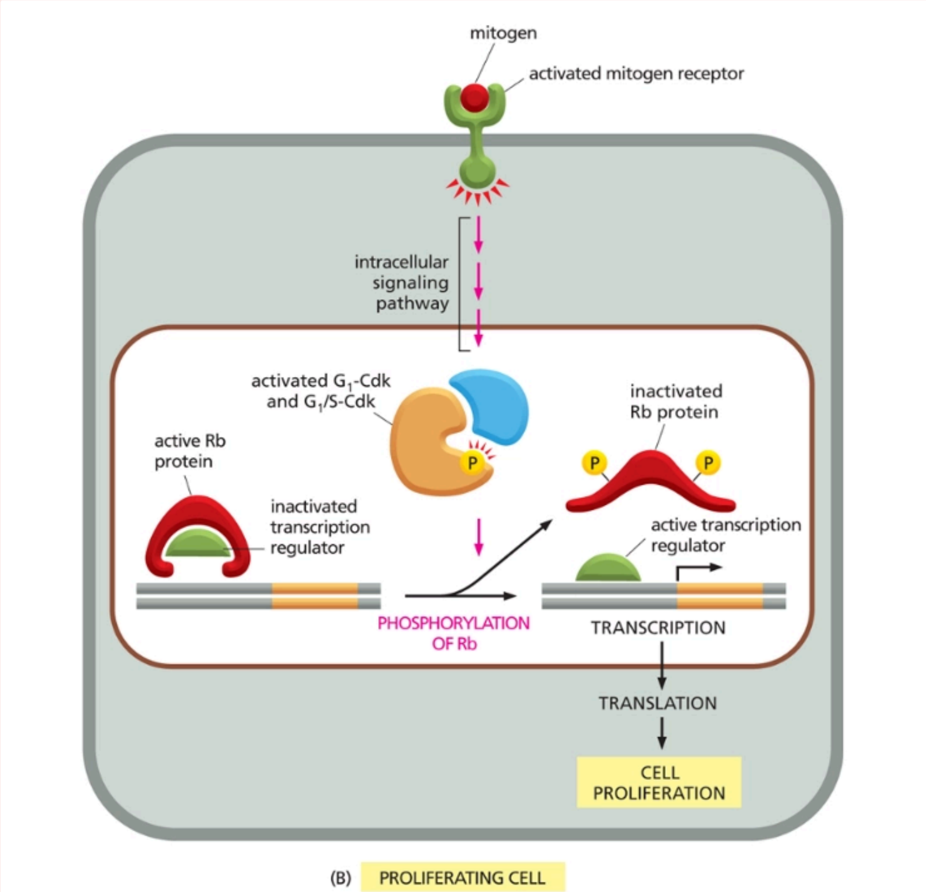
- It is a **protein** that is involved in the **regulation of the progression** of the cell cycle
- It is **found in all nucleated cells** and it acts as a **tumour suppressor**

▼ How do retinoblastomas work?

- They work by **binding onto transcription factors** and acting as a “brake” in the cell cycle.
- This is because the transcription factors **cannot turn on the genes required for cell cycle progression**
  - For example, DNA polymerase and thymidine kinase.
- This is required when the cell is “resting” and doesn’t need to proliferate

▼ How can the "brake" caused by retinoblastomas be released?

- **Activated (G1/S) CDKs** can **phosphorylate retinoblastomas** which causes them to **change in shape**, and **release the bound E2F transcription factors**
  - This means target genes such as thymidine kinase and DNA polymerase are produced by transcription factor activation.



▼ What is the E2F family?

- It describes the **family of transcription factors** which are involved in